UGLY-SPACE

On the Systematic Underexploration of
Mathematically Unattractive Solution Spaces

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October 27, 2026

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This manuscript is not peer reviewed.

Abstract

We introduce and formalize **UGLY-SPACE**, the class of mathematical problems that only admit solutions that are aesthetically unacceptable. An aesthetic validator rejects otherwise correct solutions whose ugliness exceeds a prescribed threshold. We conjecture that many long-standing open problems belong to UGLY-SPACE, explaining their historical resistance to human solutions. We interpret recent AI-generated results on k-SAT and integer multiplication as incursions into this space [3, 4], driven by search unconstrained by human aesthetic preferences. We discuss implications for mathematical discovery and the future employment of mathematicians.

1 Introduction

Mathematical taste favors some combinations of brevity, symmetry, conceptual unity, and the absence of a 47-page appendix of exceptions. The *Aesthetic Regularization Hypothesis* is that such preferences influence which candidates researchers examine. We ask what happens when a search policy charges a candidate for being unpleasant before checking whether it works.

The answer is conditional. If valid candidates are systematically assigned higher aesthetic cost, a penalty can suppress discovery. If valid candidates are cheaper, the same penalty can help. We prove these statements in a finite independent-sampling model. The resulting policy can maintain extremely high standards for results it never discovers.

We classify problems by the minimum aesthetic cost of their valid solutions. The resulting class, UGLY-SPACE, is parameterized by a representation, a verifier, and an aesthetic threshold. A budget-relative variant classifies the discovery gap between search policies with different aesthetic penalties. This separates the ugliness of a solution from the institutional willingness to encounter it.

OpenAI's October 2026 release of mathematical manuscripts provides a contemporary setting for this classification [5]. The UGLY-SPACE explanation for this wave of results is that automated search can continue into regions where a human researcher would pause to seek a more conceptual argument. We examine two concrete cases in Section 5.

2 Candidates, correctness, and taste

Fix a problem p, a nonempty finite candidate set C_p, and a verifier $V_p : C_p \to \{0, 1\}$. Let

$$S_p = \{c \in C_p : V_p(c) = 1\}.$$

We assume $S_p \neq \varnothing$. These candidates may represent proposed proofs, algorithms, or constructions, but the finite model does not presume that all possible solutions have been enumerated. A failed verification uses the same unit of budget as a successful one.

Definition 2.1 (Aesthetic cost)**.** *An aesthetic cost is a specified function $A_p : C_p \to \mathbb{R}_{\geq 0}$. It applies to invalid as well as valid candidates and is available to the sampling policy before verification. One possible schematic score is*

$$A_p(c) = w_1 L(c) + w_2 K(c) + w_3 D(c), \qquad w_i \geq 0,$$

where L is description length in a fixed language, K counts exceptional cases, and D is a declared measure of asymmetry. The weights specify the exchange rate between brevity, uniformity, and symmetry.

The language, community, and weights are part of the specification. Merely calling a construction "natural" is not a reproducible measurement, even when said by a very senior mathematician. The score is not defined by correctness; a correlation with correctness is a separate assumption that can be imposed in a toy instance or tested in an actual study.

Definition 2.2 (UGLY-SPACE)**.** *For a fixed candidate representation, verifiers V_p, aesthetic costs A_p, and threshold $\tau \geq 0$, define*

$$\mathsf{UGLY\text{-}SPACE}_\tau^{\mathrm{all}} = \{p : S_p \neq \varnothing \text{ and } \min_{s \in S_p} A_p(s) > \tau\}.$$

A problem belongs to this class precisely when it admits a valid solution and every valid candidate in its specified representation is ugly.

Definition 2.3 (Matched aesthetic samplers)**.** *Let $q_0(c \mid p) > 0$ be a probability distribution on C_p. For a finite penalty $\lambda \geq 0$, define*

$$q_\lambda(c \mid p) = \frac{q_0(c \mid p) e^{-\lambda A_p(c)}}{Z_p(\lambda)}, \qquad Z_p(\lambda) = \sum_{d \in C_p} q_0(d \mid p) e^{-\lambda A_p(d)}. \tag{1}$$

The sampler draws candidates independently with replacement and verifies each one, stopping at the first success. Write T_λ for its number of verifications, including the successful verification.

Both policies use the same verifier, candidate set, base distribution, and per-draw budget. Only λ changes. We assume exact sampling from (1) and count verifier calls; generation and scoring costs are omitted.

For the human–machine comparison set $H = q_{\lambda_H}$ and $M = q_{\lambda_M}$ with $\lambda_H > \lambda_M \geq 0$. These labels designate policies. Either policy can be implemented in carbon or silicon; the relevant parameter is its tolerance for candidates that should not be shown at a seminar. An AI system can inherit aesthetic preferences from training, while a human can run an exhaustive search after sufficient coffee.

Definition 2.4 (Budget-relative UGLY-SPACE)**.** *For a common integer budget $B \geq 1$ and $0 < \epsilon < \delta < 1$, define*

$$\textsf{UGLY-SPACE}^{H,M}_{B,\epsilon,\delta} = \{p : \mathbb{P}(T_{\lambda_H} \leq B) < \epsilon, \\ \mathbb{P}(T_{\lambda_M} \leq B) > \delta\}.$$

The universe of problems, candidate sets, verifiers, scores, and base distributions are fixed as part of this definition.

Matching the policies isolates the aesthetic penalty as the intervention. Unsolvable instances are excluded to avoid awarding the institution's aesthetic preferences credit for logical impossibility.

Definition 2.2 concerns all valid candidates in the fixed representation; Definition 2.4 concerns their discovery under specified search policies. Membership in the former is neither required by nor sufficient for membership in the latter: sampling mass and budget also matter.

3 Search under aesthetic regularization

Suppress p in notation, and put $r_\lambda = q_\lambda(S)$, the single-draw success probability. Finiteness, full support, and nonempty S imply $r_\lambda > 0$ for finite λ.

Proposition 3.1 (Discovery law)**.** *For $B \geq 1$,*

$$\mathbb{E}T_\lambda = \frac{1}{r_\lambda}, \qquad \mathbb{P}(T_\lambda \leq B) = 1 - (1 - r_\lambda)^B.$$

Proof. Independence gives $\mathbb{P}(T_\lambda > t) = (1 - r_\lambda)^t$ for integer $t \geq 0$. Summing the tail yields $1/r_\lambda$, and taking the complement of the tail at B yields the budget formula. These identities also hold when $r_\lambda = 1$. □

Proposition 3.2 (The price of taste)**.** *For the distribution in (1),*

$$\frac{d}{d\lambda} \log \mathbb{E}T_\lambda = \mathbb{E}_{q_\lambda}[A \mid S] - \mathbb{E}_{q_\lambda}[A]. \tag{2}$$

If $C \setminus S \neq \varnothing$, this also equals

$$(1 - r_\lambda)(\mathbb{E}_{q_\lambda}[A \mid S] - \mathbb{E}_{q_\lambda}[A \mid C \setminus S]).$$

Thus expected discovery time increases locally exactly when valid candidates have greater mean cost than candidates overall.

Proof. Let $W_S(\lambda) = \sum_{s \in S} q_0(s) e^{-\lambda A(s)}$. Then $\mathbb{E}T_\lambda = Z(\lambda)/W_S(\lambda)$. Differentiating its logarithm gives $Z'/Z - W_S'/W_S = -\mathbb{E}[A] + \mathbb{E}[A \mid S]$. The second identity follows by conditioning on S. If $S = C$, both terms of (2) agree and $T_\lambda = 1$ almost surely. □

The sign is determined by the alignment of aesthetic cost with validity.

3.1 A family with exponentially expensive standards

For every integer $n \geq 1$, take $C_n = \{b_1, \ldots, b_n, u\}$ with uniform q_0. Set $V(b_i) = 0$, $A(b_i) = 0$, $V(u) = 1$, and $A(u) = n$. The b_i are beautiful failures. The candidate u is correct, but looks like it was assembled during a printer malfunction.

Theorem 3.3 (Fundamental Theorem of Mathematical Ugliness)**.** *For the stated family and fixed $\lambda > 0$,*

$$r_0 = \frac{1}{n+1}, \qquad r_\lambda = \frac{1}{1 + ne^{\lambda n}}, \tag{3}$$

$$\mathbb{E}T_0 = n + 1, \qquad \mathbb{E}T_\lambda = 1 + ne^{\lambda n}. \tag{4}$$

Consequently

$$\frac{\mathbb{E}T_\lambda}{\mathbb{E}T_0} = \frac{1 + ne^{\lambda n}}{n+1} \geq \frac{1}{2}e^{\lambda n}. \tag{5}$$

For the common budget $B = n + 1$,

$$\mathbb{P}(T_0 \leq B) \geq 1 - e^{-1}, \qquad \mathbb{P}(T_\lambda \leq B) \leq 2e^{-\lambda n}. \tag{6}$$

Proof. The common factor $1/(n+1)$ cancels from (1). The invalid candidates have weight 1 each; the valid candidate has weight $e^{-\lambda n}$. Thus $r_\lambda = e^{-\lambda n}/(n + e^{-\lambda n})$, which gives (3) and (4) by Proposition 3.1. Since $n/(n+1) \geq 1/2$, (5) follows.

For the unpenalized sampler, $\mathbb{P}(T_0 > B) = (1 - 1/(n+1))^{n+1} \leq e^{-1}$, using $1 - x \leq e^{-x}$. For the penalized sampler the union bound gives

$$\mathbb{P}(T_\lambda \leq B) \leq Br_\lambda \leq \frac{n+1}{n}e^{-\lambda n} \leq 2e^{-\lambda n}.$$

The bound may exceed 1 at small n, in which case it is simply loose. □

Corollary 3.4 (The Mathematician's Curse)**.** *Fix $\lambda > 0$, $\epsilon = 0.1$, and $\delta = 0.6$. For every integer $n > \log(20)/\lambda$, the instance above belongs to* $\mathsf{UGLY\text{-}SPACE}^{q_\lambda, q_0}_{n+1,\epsilon,\delta}$.

Proof. Equation (6) gives the strict upper bound 0.1 and the lower bound $1 - e^{-1} > 0.6$ required by Definition 2.4. □

The correct candidate remains unexplored because the search policy finds it embarrassing.

Remark 3.5 (What the separation counts). The parameter n counts decoys and the assigned ugliness gap. The exponential ratio compares two fixed, memoryless policies with replacement in verifier calls. An algorithm exploiting the instance's description could simply test u; enumeration without replacement uses at most $n + 1$ verifications. Input-bit complexity and optimal search are separate questions.

3.2 How much of the result was hidden in the definition?

Proposition 3.6 (Bounded ugliness, bounded damage)**.** *If all candidates have costs in $[a, a + \Delta]$, then for $\lambda \geq 0$,*

$$e^{-\lambda\Delta} \leq \frac{\mathbb{E}T_\lambda}{\mathbb{E}T_0} \leq e^{\lambda\Delta}.$$

Proof. Using the notation in Proposition 3.2,

$$\frac{\mathbb{E}T_\lambda}{\mathbb{E}T_0} = \frac{\mathbb{E}_{q_0}[e^{-\lambda A}]}{\mathbb{E}_{q_0}[e^{-\lambda A} \mid S]}.$$

Both expectations lie between $e^{-\lambda(a+\Delta)}$ and $e^{-\lambda a}$. Taking their possible ratios proves the bounds. □

Our exponential construction uses $\Delta = n$. If costs were normalized to $[0, 1]$ while λ stayed fixed, the ratio would be at most e^{λ}, independent of n. Only the product of penalty and cost differences matters; adding a constant to all costs changes no policy, and rescaling costs can be offset by rescaling λ.

For a reverse example, let the single valid candidate have cost 0 and the n invalid candidates have cost n. Then

$$\mathbb{E}T_\lambda = 1 + ne^{-\lambda n} \leq n + 1 = \mathbb{E}T_0.$$

Beauty is now an excellent heuristic. We decline to retract the title.

Finally, a *hard* aesthetic cutoff is a different policy: condition q_0 on $\{c : A(c) \leq \tau\}$. If that set is nonempty and excludes all valid candidates, success probability is zero and discovery time is infinite. If the set is empty the conditioned policy is undefined. Finite soft penalties do not eliminate any candidate under our full-support assumption, and hence always have finite mean discovery time.

4 Numerical illustration

Table 1 evaluates the theorem's formulas for $\lambda = 0.5$ and $B = n + 1$. The dependency-free script `experiments/toy_model.py` generates this table and a CSV by deterministic evaluation of the formulas.

n	$\mathbb{E}T_M$	$\mathbb{E}T_H$	Ratio	$\mathbb{P}(T_M \leq B)$	$\mathbb{P}(T_H \leq B)$
1	2	2.65	1.32	0.750000	0.612544
4	5	30.56	6.11	0.672320	0.153267
8	9	437.79	48.64	0.653561	0.020371
16	17	47,696.33	2,805.67	0.643214	0.000356

Table 1: Constructed ugly-solution instances. $H = q_{0.5}$ and $M = q_0$. Times count verifier calls; success columns use $B = n + 1$. Values are rounded.

The accompanying checks independently enumerate small candidate sets and compare their normalized weights with the closed forms. They also check the derivative, the reverse example, the bounded-range estimate, and zero-penalty and all-valid boundary cases.

5 Recent encounters with UGLY-SPACE

The recent supply of AI-generated mathematical arguments invites a classification by aesthetic cost. Two OpenAI preprints are particularly instructive: one replaces an exponential-time hypothesis with a complicated subexponential bound; the other improves an exponent by an amount that requires considerable typesetting discipline to distinguish from zero.

Exact counting for k-SAT. In *The Exponential Time Hypothesis is False*, OpenAI reports a deterministic algorithm that exactly counts satisfying assignments of a k-CNF with n variables and m clauses, for each fixed $k \geq 2$, in polynomial space and time

$$(n+m)^{O_k(1)} 2^{O_k\left(n \log\log n/\sqrt{\log n}\right)}.$$

Theorem 1.1 of the preprint gives this bound, with a sharper exponent at fixed clause density [3]. The reported counting algorithm also decides satisfiability, giving the stated refutation of ETH. Its construction passes through shared linear summaries, sparse polynomial zero counting, and tensor contraction; Appendix D supplies the sharper counting bound. The hypothesis is removed, while a substantial amount of exposition remains available for human consumption. In our classification, the relevant aesthetic costs include the layered construction and the distance between the simplicity of the question and the form of the answer.

Integer multiplication below $n \log n$. A second OpenAI preprint reports exact deterministic multiplication of two n-bit integers on a fixed finite-tape Turing machine in time

$$O\Big(n(\lg n)^{1-\kappa}\Big), \qquad \kappa = 2^{-182},$$

where lg is the base-two logarithm with the paper's rounding convention at small inputs [4]. The manuscript describes its constants and thresholds as extremely large. Ignoring hidden constants, the formal saving factor $(\lg n)^{\kappa}$ reaches two only when

$$\log_2 \log_2 n = 2^{182}.$$

The asymptotic improvement has been secured; arranging a demonstration is left to experimental colleagues. This example places a small numerical constant, a large construction, and a distant asymptotic regime in the same proof, allowing several components of aesthetic cost to be charged at once.

Interpretation. The UGLY-SPACE account attributes such discoveries to a lower aesthetic penalty during candidate selection. In the model, λ_M near zero allows a machine to pursue a formally promising construction without first obtaining agreement that it is natural. The resulting division of labor is immediate: the machine supplies the argument, and humans explain why a better argument ought to exist. For empirical purposes, these cases are candidates for the budget-relative classification; their reported bounds and the aesthetic interpretation play distinct roles.

6 Conjectures

UGLY-SPACE Conjecture. Many longstanding open problems have resisted human solution partly because their accessible solutions lie in regions of candidate space systematically neglected for aesthetic reasons. AI-assisted search may resolve such problems by exploring those regions with less aesthetic inhibition. In the terminology of Definition 2.4, the conjecture proposes that UGLY-SPACE has substantial real-world membership under suitable budgets and matched policies. The empirical task is to estimate the size of this family while controlling for compute, representation, and verification costs.

Aesthetic Regularization Hypothesis. In some mathematical research settings, aesthetic judgments influence candidate selection beyond their contribution to expected correctness or verification cost. Testing this would require an operational score, observed search behavior, and control for expertise, resources, and verification difficulty.

Conjecture (Grothendieck–GPU Tradeoff). In some settings, increasing preference for elegance improves the elegance of accepted results while reducing the fraction of solvable problems solved within budget. The direction of the tradeoff should depend on adaptive exploration, reuse of abstractions, and verification costs.

Optimist's Conjecture. Fix a proof system, language, nonnegative aesthetic score A, and threshold τ. For each provable statement p, let $\Pi(p)$ be its nonempty set of finite proofs. The optimist proposes

$$\forall p \text{ provable}, \quad \exists \pi \in \Pi(p) : A(\pi) \leq \tau.$$

Every ugly proof would then have an elegant alternative, though neither a method nor a budget for finding it is promised.

Pessimist's Conjecture. The pessimist proposes the negation:

$$\exists p \text{ provable}, \quad \forall \pi \in \Pi(p) : A(\pi) > \tau.$$

Both conjectures concern the same fixed proof system, language, score, and threshold. Changing these choices can change the answer. The pessimist's formula is shorter and therefore aesthetically preferable.

7 Related work

The exponential weights in (1) use the familiar statistical mechanics form. Kirkpatrick, Gelatt, and Vecchi discuss a connection between statistical mechanics and optimization [2]. We use independent draws from a fixed distribution, with the aesthetic penalty playing the role of inverse temperature. A sufficiently refined community is consequently at risk of freezing before it has found anything.

The formal proof of the Kepler conjecture by Hales and collaborators is a substantial proof checked using HOL Light and Isabelle [1]. Such verification separates the correctness of a proof from the resources required to inspect it. Our model abstracts this distinction through a verifier and a separate candidate-selection policy.

8 Implications for the future employment of mathematicians

If machines became willing to inspect all aesthetically disreputable candidates, mathematicians could specialize in explaining why the resulting proofs ought not to have been necessary. This suggests three emerging occupations: proof beautician, exception-list curator, and senior vice president for conceptual inevitability.

Choosing worthwhile questions, discovering representations, teaching, and building verifiers would remain substantial occupations. The present model prices neither these activities nor electricity. We therefore reserve salary estimates for a more complete theory, or for the keynote.

A division of labor follows: automated exploration can offer candidates, formal tools can check them, and people can develop explanations and useful abstractions. The efficiency of this arrangement remains to be measured. Our contribution is a table with four rows and an alarming degree of confidence in its typography.

9 Limitations

The construction deliberately correlates high aesthetic cost with validity. The discovery-time separation depends on that alignment and on the growing aesthetic range. Independent sampling forgets failures; real search can learn, cache, adapt, and share results. The finite candidate sets omit unbounded proof search, and the resource model omits candidate-generation, scoring, and variable verification costs. Costs are subjective and representation dependent. Applying the model to research practice requires operational aesthetic scores and matched search comparisons. The examples in Section 5 concern particular reported solutions; establishing membership in $\mathsf{UGLY\text{-}SPACE}_{\tau}^{\mathrm{all}}$ would require a lower bound on the aesthetic cost of every valid solution in the fixed representation. Learned search policies can also inherit aesthetic preferences from their training data.

The paper itself may constitute empirical evidence for UGLY-SPACE. A control group was considered but rejected on aesthetic grounds.

10 Conclusion

Aesthetic penalties can exponentially delay discovery in a deliberately constructed finite sampling family. The sign of the effect depends on how cost aligns with validity, and its magnitude depends on the penalty range. The UGLY-SPACE Conjecture proposes that this mechanism matters for a substantial family of open problems. We recommend occasionally checking whether the hideous candidate is correct before commissioning a nicer one.

References

[1] Thomas Hales, Mark Adams, Gertrud Bauer, Tat Dat Dang, John Harrison, Le Truong Hoang, Cezary Kaliszyk, Victor Magron, Sean McLaughlin, Tat Thang Nguyen, Quang Truong Nguyen, Tobias Nipkow, Steven Obua, Joseph Pleso, Jason Rute, Alexey Solovyev, Thi Hoai An Ta, Nam Trung Tran, Thi Diep Trieu, Josef Urban, Ky Vu, and Roland Zumkeller. A formal proof of the Kepler conjecture. *Forum of Mathematics, Pi*, 5:e2, 2017. https://doi.org/10.1017/fmp.2017.1.

[2] S. Kirkpatrick, C. D. Gelatt, Jr., and M. P. Vecchi. Optimization by simulated annealing. *Science*, 220(4598):671–680, 1983. https://doi.org/10.1126/science.220.4598.671.

[3] OpenAI. The exponential time hypothesis is false. OpenAI Math Release preprint, October 17, revised October 25, October 2026. Theorem 1.1 and Appendix D. Source version inspected October 27, 2026.

[4] OpenAI. Integer multiplication below $n \log n$. OpenAI Math Release preprint, September 23, September 2026. Theorem 1. Source version inspected October 27, 2026.

[5] OpenAI. Sharing AI progress in mathematics. Research announcement, October 6, October 2026. Official announcement.